NTNU

Department of computer science

Examination paper for TDT4175 Information Systems

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Examination time (from-to): 9:00-12:00

Permitted examination support material: *Allowed aids: C Specified printed and handwritten material is allowed. In the run of the course the book* "BPMN Method & Style", Bruce Silver. (Edition 1 or 2) has been used. Clean copies of the chapters 1-6 used in the course or of the summary text put out on Black Board are also allowed. *In addition standard dictionaries (English-English, and from English to other languages) are allowed to bring. No additional printed or handwritten materials are allowed. An approved simple calculator is allowed.*

Language: English

Number of pages (front page excluded): 3

Number of pages enclosed: 4

Informasjon om trykking av eksamensoppgave

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Checked by:

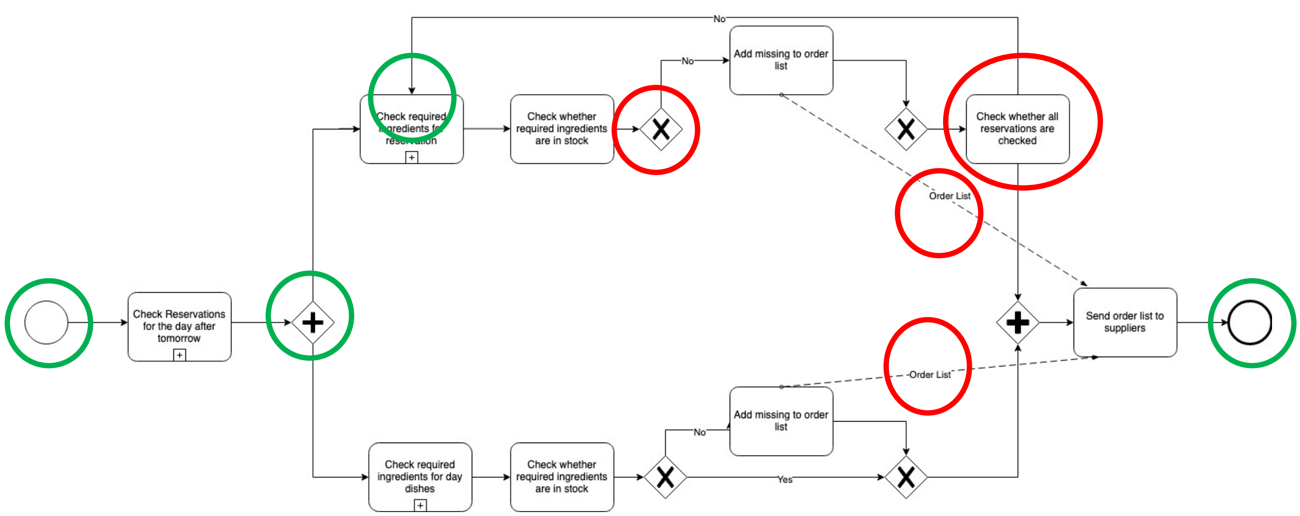
Nana Kwame Henebeng Amagyei

Date Signature

Task 1 - (25 points)

Julie and Eva have opened a new restaurant with the name 'Dine for Life', which is open every day. They aim to serve climate friendly food (vegetarian and bought from local suppliers) and to minimize food waste. To achieve this last objective, they want to buy in just enough ingredients to cover their needs. Customers can come to the restaurant without reservation, but then they have only one choice of food: the dish of the day. They can also make a reservation, at least 2 working days before they want to have dinner. Along with the reservation they must choose what to eat from the menu. When a reservation is made, it is checked whether there is an empty table for the required number of people. If so, the table is reserved. Customers must confirm their visit a day before the reservation date. If no confirmation is received, the reservation is cancelled. At the end of every working day, the list with reservations for two days later is checked. Based on the required ingredients for the ordered food per reservation, the list of required ingredients for the reservation day is updated. All ingredients that are not in stock are ordered from the local suppliers. The deal is that orders are delivered the next day unless that is a Sunday. In that case the orders are delivered on Monday. Tables not reserved at a given day are open for drop-inn customers. There must be enough ingredients to serve all the not reserved places the dish of the day. Ingredients needed for this are added to the required ingredients list.

Given this case description a student has made the following model.



1.1 (5 points)
The first step of modelling a business process, as advocated by Silver, is choosing the scope of your model. One of the questions to be answered in scoping is: **What does each instance of the process represent?** Answer this question for the model made by the student.

An instance of the process represents compiling and sending the list of ingredients to be ordered.

1.2 (20 points)
The model contains several mistakes, both syntactical and semantical. Find at least three syntactical and two semantical mistakes (so, five mistakes in total). Motivate why you consider these to be mistakes.

Semantical mistakes: The start event should be a timer start event because the process starts every evening after the restaurant is closed. The AND split is wrong. You have to do the planning for the non-reserved tables after the planning for the reserved tables is ready, otherwise you don't know how many empty tables you have. The loop to 'Check required ingredients for reservation' is wrong, should go to 'Check reservations for the day after tomorrow. The end event should be a message end event because it generates a message to the suppliers.

Syntactical mistakes: the XOR gateway lacks an outgoing arrow. Message flows are not allowed within a white box. The activity 'check whether all reservations are checked' must be followed by an XOR gate.

Task 2 (5 points)

What is a computer-based information system (CBIS)?

Computer-based information system is a single set of hardware, software, databases, telecommunications, people, and procedures that are configured to collect, manipulate, store, and process data into information.

Task 3 (5 points)

Describe the difference between data, information and knowledge using examples from the restaurant case.

Data consist of raw facts, such as an employee number, total hours worked, number of reservations, number of unreserved tables. When facts are arranged in a meaningful manner they become information, i.e. information is a collection of facts organized in such a way that they have value beyond the facts themselves. Examples from the case can be: number of hours worked by employee Annie in July 2022, number of reservations for October 12, 2022, number of rissoto's ordered for October 12, 2022. Knowledge is understanding of a set of information and using this understanding in acting. Examples from the case can be: understanding which ingredients are needed given the number of reserved dishes at a given date, understanding how many meals of the day must be produced given the number of unreserved tables at a given day. Other examples are possible also, but they should exemplify the difference between data, information and knowledge.

Task 4 (15 points)

Eva and Julie notice that compiling the order list at the end of every working day is a lot of work when done manually. They work late hours and often the list contains errors because they make mistakes due to being tired. A friend suggests automating (part of) this activity by using an information system. Which type of information system would be suitable to use here? Describe the functionality of the system required to support the activity.

The most appropriate system here is a decision support system. It could help in selecting the required ingredients per ordered dish and the required meals of the day, as well as checking what should be ordered given the ingredients in stock.

Task 5 (15 points)

After a year Eva and Julie evaluate their achievements with respect to their main business goals. Especially with respect to reducing food waste, they are disappointed. They have the feeling that they had to throw a good deal of ingredients that were unused as well as prepared food but are unsure about the reason for this. They have used a transaction management system that registers reservations, cancelations, payments, orders of ingredients, deliveries of ingredients, payments to personnel, etc.

5.1 (5 points)
What basic transaction processing activities are performed by all transaction processing systems?

Data collection; Data editing; Data correction; Data manipulation; Data storage; Document production and reports.

1 point per correct activity (so they can miss one and still get 5 points)

5.2 (5 points)
Which kind of information system could be used to get insight in the possible reasons for food waste, using available transaction information?

A Management Information System (MIS)

5.3 (5 points)
Which views (such as the number of no shows per day of the week) could be useful to find out the reason for having to throw food?

Here are there several possibilities. Examples:

- Number of unused meals of the day per day of the week
- Amount of ingredients that had to be thrown away because they were overdue per month
- Deviations between orders and deliveries over a month
- Failures in food preparation over a month
- …

Students can be creative here. The point is that they show how aggregated reports can give insight.

Task 6 (15 points)

The order lists are sent till local suppliers by e-post. This has led to delays and misunderstandings (wrong deliveries, for example). Because of these uncertainties the two days ordering window was chosen. For some goods (fresh goods that don’t last that long) deliverance at the day of use would be better.

6.1 (7 points)
Which kind of information system could be used to improve the order/delivery process?

A Supply Chain Management System (SCMS)

6.2 (8 points)
Propose a good way to reorganize the ordering/delivery process using the type of system mentioned in 6.1.

The point here is that the system should integrate the order and delivery process. There are different levels at which this can be achieved. The most basic level is that once the order list is compiled it is automatically sent to the different local suppliers. So, the SCMS keeps track of which ingredients should be ordered from which supplier and sends the orders. The most advance form is that the SCMS has access to the required ingredients, the ingredients in stock (along with their due date) and generates orders for the respective suppliers. There are other possibilities also. Most important is that the student understands that the reorganization aims at integrating ordering and delivery processes.